

ADITI GUPTA

Applied Machine Learning | Electronics & Telecommunication Engineering
Pune, Maharashtra – 411043 | [+91-7805999464](tel:+91-7805999464) | ag0484363@gmail.com

PROFESSIONAL SUMMARY

Electronics & Telecommunication Engineering graduate with a DRDO research background and a Best Paper Award for co-authored work on responsible, risk-aware AI. Applied machine learning experience includes CNN-based multimodal sensor classification, alongside full-stack engineering and data analytics. Seeking to bring this cross-disciplinary foundation to an applied AI/ML research role.

TECHNICAL SKILLS

Machine Learning: CNN-based multimodal sensor classification, applied deep learning fundamentals, digital signal processing (FIR/IIR filter design)

Languages: Python, Java, C, JavaScript, TypeScript, SQL

Backend & APIs: Spring Boot, Node.js, RESTful API design, MVC architecture

Data & Analytics: SQL, MongoDB, Power BI, Tableau, Chart.js

Cloud & DevOps: GCP, AWS, Git, CI/CD fundamentals

Core CS & Tools: DSA, OOP, DBMS, Agile/Scrum, GitHub, VS Code, LaTeX, Jupyter

EXPERIENCE

Team Member — IDRO (Research & Innovation Foundation)

Jul 2025 – Present

- Collaborated cross-functionally to design IDRO's GitHub organization page, establishing the foundation's open-source presence and developer-facing documentation.
- Contributed to IDRO's first indigenous drone project end-to-end — from system design through 3D-printed assembly to a flight-ready prototype — in an agile, iterative development cycle.

Research Intern — Defence Institute of Advanced Technology (DIAT), DU, DRDO, Pune

May 2025 – Jul 2025

- Co-authored a research paper on RFID antenna design, using CST Studio Suite for electromagnetic simulation, modeling, and validation, delivering research output within a 3-month R&D cycle.

KEY PROJECTS

SomniSense — Multimodal Sleep-Stage Classification System

CNN, Sensor Fusion, ESP32/FreeRTOS, MQTT, MongoDB

- Designed a patent-oriented hybrid sleep monitoring system fusing contact biosensors (MAX30102, GSR, TMP117, BNO055) with RD-03E mmWave radar for non-contact sensing.
- Built a CNN-based classification pipeline for sleep-stage detection, with adaptive mode-switching between sensor modalities to maintain reliability when contact sensors are dislodged.
- Developed the end-to-end system architecture, including custom PCB design in EasyEDA, embedded firmware, and a cloud backend for data ingestion and storage.

Financial Behavior Analysis Platform (Expense Tracker)

JavaScript, Spring Boot, SQL

- Engineered a full-stack expense management system using Spring Boot (MVC) and RESTful APIs to automate transaction tracking, categorization, and data persistence.
- Built an interactive Chart.js dashboard visualizing spending trends to support faster, data-driven financial insights.

Healthcare Data Analytics & Decision Support (Hospital Management System)

SQL, Power BI

- Architected a normalized relational schema for multi-entity hospital data with automated billing logic to streamline operational workflows.
- Developed KPI-driven Power BI dashboards for real-time reporting, trend analysis, and anomaly detection.

Digital Hearing Aid

Python, DSP

- Built a real-time signal-processing pipeline comparing FIR vs. IIR filter architectures for noise reduction, benchmarked at sub-100ms latency.

PUBLICATIONS & AWARDS

- Best Paper Award (Co-Author) — RRAI 2026 International Conference on Responsible, Risk-Aware, and Regulated AI, BV(DU)COEP — recognized for methodological rigor in AI safety, algorithmic transparency, and bias mitigation.

EDUCATION

B.Tech, Electronics & Telecommunication Engineering — Bharati Vidyapeeth (Deemed to be University) COE, Pune

2022 – 2026 | CGPA: 8.4

CERTIFICATIONS & LEADERSHIP

- 100 Days of Code: The Complete Python Pro Bootcamp — Udemy.
- Java Masterclass 2025 (130+ hrs) — Udemy.
- Editorial & Envoy Head, ETSA — BV(DU)COEP: Led a cross-functional content team through strategy and production, publishing 4 magazine editions (Sep 2024–Aug 2025).
- Design Team Member, GDSC — BV(DU)COEP: Delivered UI/UX and visual design for a 300+ member developer community (Sep 2023–Jun 2024).